

Technology Stack

Date	14 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

The Personalized Networking Assistant uses a modern full-stack architecture built with React, Vite, Tailwind CSS, Node.js, Express.js, SQLite3, Google GenAI and Wikipedia APIs. The selected technologies support an interactive web experience, server-side AI processing, persistent local data storage, factual context retrieval and dashboard visualizations.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React 18+, Vite, Tailwind CSS, motion/react, Lucide React, Recharts	Provides a modern component-based interface, fast Vite development and bundling, responsive Tailwind styling, interactive animations, vector icons, and dashboard data visualizations.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Node.js 18/20 LTS, Express.js, tsx, esbuild	Handles API routing, CORS, static file serving and JSON requests. tsx supports development execution, while esbuild bundles the production server. A custom console logger records server information, warnings and errors.

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3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	SQLite3	Uses a self-contained, file-based relational database with zero network configuration. The active application database is stored locally as <code>networking_assistant.db</code> .
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Not specified in Technology Stack.md	The authoritative Technology Stack Markdown does not specify a cloud hosting or deployment platform, so no unsupported platform is added here.

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5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub	Hosts and manages the project source repository and documentation. No additional CI/CD technology is specified in the Technology Stack Markdown.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	@google/genai (Gemini 3.5 Flash), Wikipedia REST & Action APIs	The server-side Google GenAI SDK provides the core AI capability using Gemini 3.5 Flash, while Wikipedia REST and Action APIs provide external knowledge access for factual context.